

DEPARTMENT OF BIOTECHNOLOGY / ENGINEERING SCIENCES

BIOLOGY FOR ENGINEERS

(B.Tech — Second Year, Semester III/IV)

COMPREHENSIVE STUDY MATERIAL

Unit: Cell Biology — Cell Organelles, Structure & Function

Prepared for classroom instruction and self-study

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Preface — A Note from the Faculty

This study material has been compiled to support second-year B.Tech students studying the foundational unit on Cell Biology within the Biology for Engineers curriculum. Over twenty-five years of teaching this subject to engineering cohorts, I have found that students grasp cellular concepts best when the material bridges classical biology with engineering intuition — treating the cell as a highly optimised, self-regulating micro-system with membranes as selective barriers, organelles as functional sub-units, and enzymes as biological catalysts governed by rate kinetics familiar from process engineering.

This document is organised into clearly demarcated units, each opening with learning outcomes and closing with a concise summary. Key terms are boldfaced on first use and consolidated in the glossary. Comparative tables are provided wherever the distinction between cell types or structures is examination-relevant. A bank of review questions, including short-answer, long-answer, and multiple-choice items with an answer key, is provided at the end for self-assessment and examination preparation.

Students at NIRF Top-50 institutes are expected not merely to memorise structures but to appreciate their functional and interdisciplinary significance — particularly in biomedical engineering, tissue engineering, drug delivery, biomaterials, and biosensor design. Throughout this material, callout boxes titled "Engineering Relevance" draw these connections explicitly.

How to use this material

Read each section once for conceptual understanding, then revisit the summary tables and diagrams for revision. Attempt the review questions without referring back to the text, and use the glossary to test recall of terminology before your examinations.

Course/Unit Learning Outcomes

On successful completion of this unit, students will be able to:

- CO1: Describe the fundamental structural organisation of the cell and distinguish between prokaryotic and eukaryotic cell types.
- CO2: Explain the structure and function of major cell organelles, including the cell membrane, cell wall, cytoplasm, nucleus, endoplasmic reticulum, and Golgi apparatus.
- CO3: Differentiate plant, animal, and bacterial cells on the basis of structural and biochemical criteria.
- CO4: Explain the hierarchical organisation of biological systems — cells, tissues, organs, and organ systems.
- CO5: Describe the structure, characteristics, and mechanism of enzyme action, and identify physiological examples.
- CO6: Explain the structure and functional role of chlorophylls and carotenoids in photosynthetic pigmentation.
- CO7: Relate cellular and biomolecular concepts to engineering applications in biomedical, tissue, and biomaterials engineering.

Unit 1: The Cell — Fundamental Unit of Life

1.1 What Is a Cell?

A cell is the structural and functional unit of life. The branch of biology devoted to the study of cells — from basic structure to the function of individual organelles — is known as Cell Biology (or Cytology). The English natural philosopher Robert Hooke is credited with the discovery of cells in 1665, when he observed thin slices of cork under a primitive compound microscope and noted small, box-like compartments that he termed "cells," after the small rooms occupied by monks.

All living organisms are composed of cells. Organisms composed of a single cell are termed unicellular (e.g., bacteria, amoeba), while those composed of many cells are termed multicellular (e.g., plants, animals, humans). Cells vary enormously in number across organisms — a single bacterium is one complete cell, whereas the human body comprises approximately 37 trillion cells. Despite this variation, cells remain the lowest level of structural and functional organisation common to every living system, and it is within cells that the hereditary material (DNA) of an organism resides.

1.2 Range of Cell Size

Cells range in size from as small as 0.1 μm in diameter (Mycoplasma, the smallest known free-living cells) to structures nearly 150 mm across — the ostrich egg being among the largest single cells known in nature. Despite this vast range, spanning roughly six orders of magnitude, all cells share a common organisational plan that enables them to carry out the essential processes of life: metabolism, growth, response to stimuli, and reproduction.

Engineering Relevance

The scale at which cellular structures operate (nanometres to micrometres) places cell biology squarely within the domain of nanotechnology and microfabrication. Biomedical and nanomaterials engineers routinely design scaffolds, drug carriers, and biosensors whose dimensions are matched to cellular and sub-cellular length scales.

1.3 Prokaryotic versus Eukaryotic Cells

All cells on Earth fall into one of two fundamental categories: prokaryotic or eukaryotic. This distinction is among the most important organising principles in biology, with direct implications for how organisms grow, reproduce, and evolve.

Feature	Prokaryotic Cells	Eukaryotic Cells
Nucleus	Absent — DNA is freely suspended in a region called the nucleoid	Present — a true, membrane-bound nucleus containing DNA
Cellularity	All single-celled (unicellular)	May be unicellular or multicellular
Examples	Archaea, bacteria, cyanobacteria	Plants, fungi, protozoans, and animals

Feature	Prokaryotic Cells	Eukaryotic Cells
Cell size	0.1 to 0.5 µm in diameter	10 to 100 µm — up to 200× larger than prokaryotes
Reproduction	Primarily binary fission (asexual); some conjugation	Both sexual and asexual reproduction
Hereditary material	Can be DNA or RNA	DNA, organised into chromosomes
Organelles	Lack membrane-bound organelles	Contain membrane-bound organelles (mitochondria, ER, Golgi, etc.)
Cell wall / chloroplasts	Cell wall present in most cases	Plant cells possess chloroplasts and a cell wall; animal cells possess neither

The evolutionary transition from prokaryotic to eukaryotic cells is believed to have occurred approximately two billion years ago through a process known as endosymbiosis, in which free-living prokaryotes were engulfed by, and became permanently integrated with, a host cell. This transition enabled the subsequent development of complex multicellular life and is regarded as one of the most significant events in the history of life on Earth.

Unit 2: Cell Structure and Organelles

The cell is composed of individual structural components, each performing specific functions essential to sustaining life. These components include the cell membrane, cell wall (in plants, fungi, and most bacteria), cytoplasm, nucleus, and a variety of specialised organelles. Each component plays a distinct and critical role in maintaining cellular integrity and function.

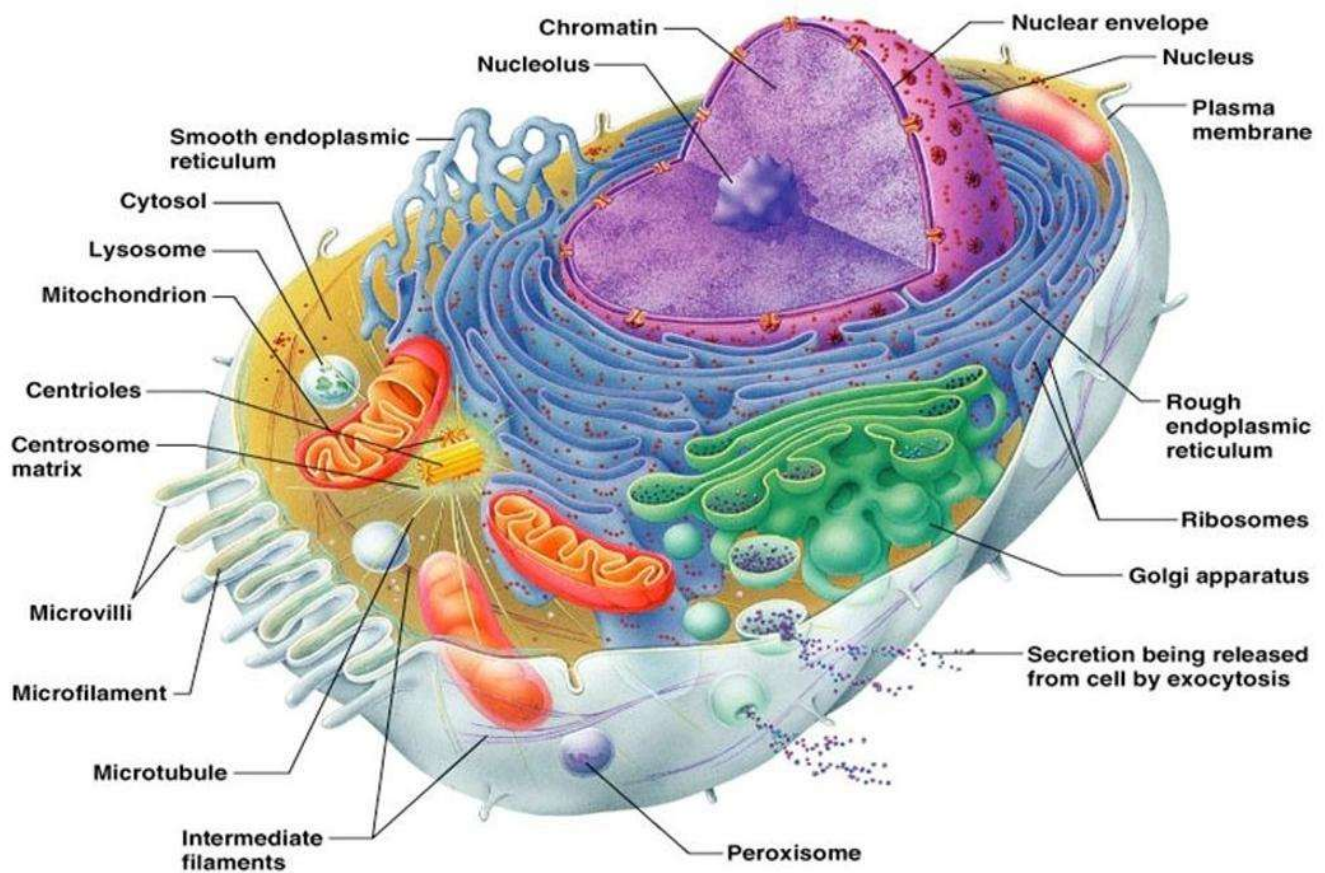


Figure 2.1: Generalised eukaryotic cell showing major organelles.

2.1 The Cell (Plasma) Membrane

The cell membrane, also called the plasma membrane, is the outer covering of the cell within which all other organelles — including the cytoplasm and nucleus — are enclosed. It is present in every cell, both prokaryotic and eukaryotic, and performs several critical functions:

- **Support and Protection:** Supports and protects the cell, separating it from the external environment and shielding internal components from damage.
- **Selective Permeability:** Structurally porous, it permits the movement of selected substances into and out of the cell — a defining property known as selective permeability.
- **Boundary Formation:** Forms the interface between adjacent cells as well as between the cell and its surroundings.

- **Communication:** Contains receptor proteins that allow the cell to receive chemical signals from hormones, neurotransmitters, and neighbouring cells.

Structurally, the cell membrane conforms to the Fluid Mosaic Model, first proposed by Singer and Nicolson: it is a dynamic phospholipid bilayer with embedded proteins that are free to move laterally within the plane of the membrane, conferring both flexibility and functional versatility.

Engineering Relevance

The selective-permeability behaviour of the plasma membrane is the biological inspiration for synthetic semi-permeable membranes used in dialysis, controlled drug-release systems, and membrane bioreactors. Understanding lipid bilayer mechanics is foundational to liposomal drug delivery design.

2.2 The Cell Wall

The cell wall is the most prominent structural feature distinguishing plant cells; it is also found, with different composition, in fungi and some bacteria. Its principal characteristics are as follows:

- **Composition:** Composed primarily of cellulose, hemicellulose, and pectin — a tough, fibrous combination that confers rigidity.
- **Location:** Forms the outermost layer of the plant cell, surrounding and protecting the plasma membrane and all internal contents.
- **Structure:** A rigid, stiff structure that preserves cell shape even under mechanical or osmotic stress.
- **Function:** Provides shape, structural support, and protection against mechanical shock, injury, and excessive osmotic pressure.
- **Permeability:** Unlike the selectively permeable cell membrane, the cell wall is fully permeable, allowing water and dissolved solutes to pass through freely.

2.3 The Cytoplasm

The cytoplasm is a thick, translucent, jelly-like substance occupying the space between the plasma membrane and the nucleus. Its principal roles include:

- **Site of Chemical Reactions:** The majority of a cell's metabolic reactions occur within the cytoplasm, including glycolysis, protein synthesis, and signal-transduction pathways.
- **Organelle Suspension:** Organelles such as the endoplasmic reticulum, vacuoles, mitochondria, and ribosomes are suspended within the cytoplasm, held in position by the cytoskeleton.
- **Composition:** Approximately 80% water, together with dissolved salts, proteins, and various organic molecules.
- **Cytoskeleton:** A network of protein filaments (actin filaments, microtubules) that provides structural support and facilitates intracellular transport.

The cytoplasm exclusive of its organelles is referred to as the cytosol; together, the cytosol and the organelles constitute the complete cytoplasm.

2.4 The Nucleus — Command Centre of the Cell

The nucleus (plural: nuclei) is the most integral component of the eukaryotic cell and is the defining feature distinguishing eukaryotes from prokaryotes, which possess only a nucleoid region rather than a true, membrane-bound nucleus. The term derives from the Latin for "kernel of a nut," an apt description of this dense, central structure.

The Nucleus

Command center of the eukaryotic cell

Chromatin

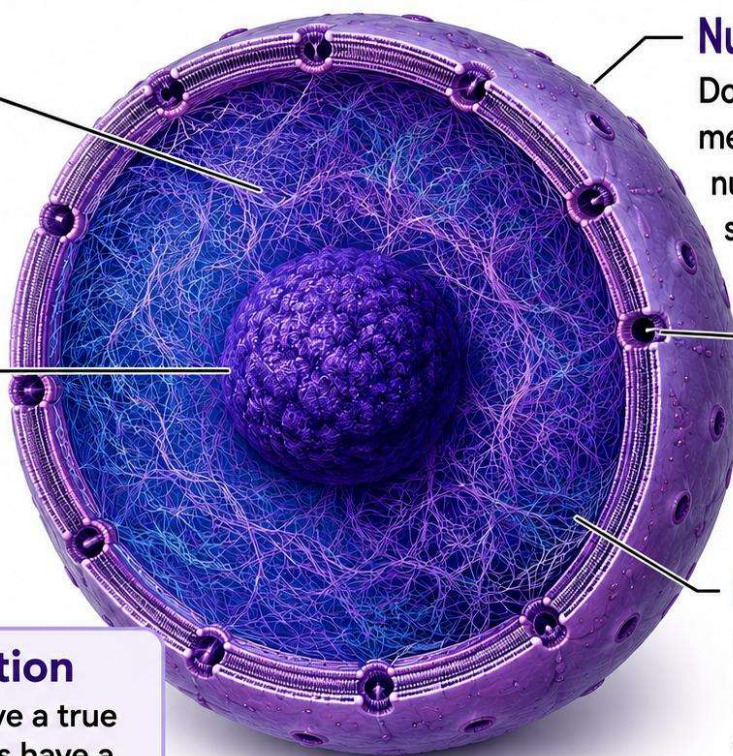
DNA wrapped around histone proteins; condenses into visible chromosomes during cell division

Nucleolus

Dense, non-membrane-bound structure where ribosomal RNA (rRNA) is synthesized and ribosome subunits are assembled

Key Distinction

Only eukaryotes have a true nucleus. Prokaryotes have a **nucleoid** — an undefined region where DNA floats freely without a surrounding membrane.



Nuclear Envelope

Double lipid bilayer membrane with nuclear pores for selective transport

Nuclear Pore

Regulates movement of molecules in and out of the nucleus

Nuclear Matrix

Protein scaffold that provides structural support and organizes chromatin

Figure 2.2: Structure of the nucleus, showing the nuclear envelope, nucleolus, and nuclear pores.

Key Functions of the Nucleus

- Stores Hereditary Material: Houses the cell's DNA — the complete genetic blueprint of the organism, organised into chromosomes.
- Controls Cell Activity: Regulates cell growth, maturation, division, and death through control of gene expression.
- Protects DNA: The nuclear envelope, a double membrane, separates DNA from the rest of the cell while allowing controlled exchange of materials via nuclear pores.

- **Ribosome Production:** The nucleolus, a dense sub-region within the nucleus, produces ribosomes, which are subsequently exported to the cytoplasm for protein synthesis.
- **Orchestrates Cell Division:** Directs the precise replication and distribution of DNA to daughter cells during mitosis or meiosis.

Quantitative Facts

- If stretched end-to-end, the DNA within a single human cell nucleus would measure approximately 6 feet (about 2 metres), despite being packed into a structure only ~6 µm in diameter.
- Each nucleus contains approximately 2,000 to 4,000 nuclear pore complexes, which regulate molecular transport between the nucleus and the cytoplasm.
- The human genome comprises approximately 3 billion base pairs of DNA, encoding roughly 20,000 to 25,000 genes.

Key Distinction

Only eukaryotes possess a true, membrane-bound nucleus. Prokaryotes instead possess a nucleoid — an undefined region in which DNA floats freely, unenclosed by a nuclear membrane. This is one of the most fundamental distinctions between the two cell types.

2.5 The Endoplasmic Reticulum (ER)

Endoplasmic Reticulum (ER)

The ER is a network of membrane-bound sacs and tubules that plays a key role in protein and lipid synthesis, folding, and transport.

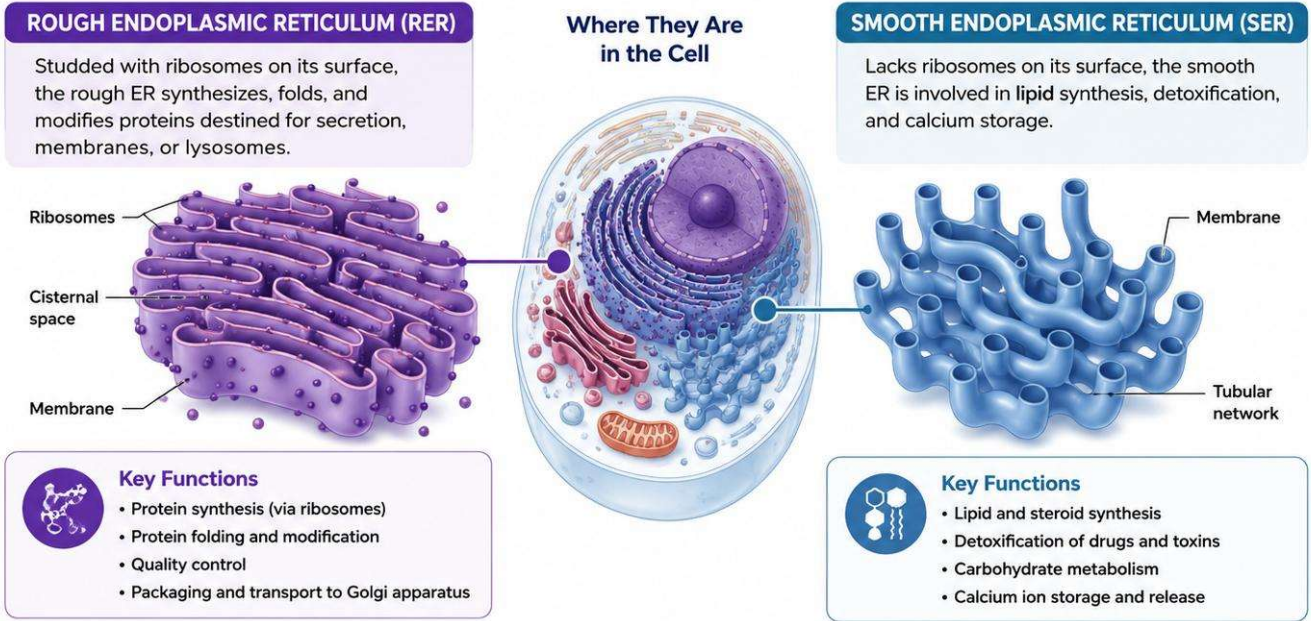


Figure 2.3: Structure of the Endoplasmic reticulum

The endoplasmic reticulum occurs in two structurally and functionally distinct forms. The Rough Endoplasmic Reticulum (RER) has a surface studded with ribosomes and is the principal site of protein synthesis and initial

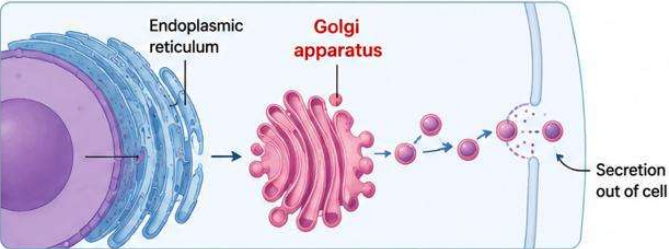
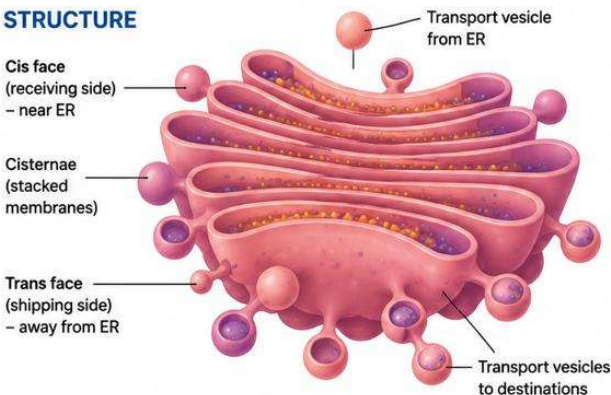
folding. The Smooth Endoplasmic Reticulum (SER) lacks ribosomes and is chiefly involved in lipid synthesis and detoxification. Both membranes form continuous folds that ultimately join the outer layer of the nuclear membrane. With the exception of sperm cells and red blood cells, the endoplasmic reticulum is present in essentially every type of eukaryotic cell.

2.6 The Golgi Apparatus

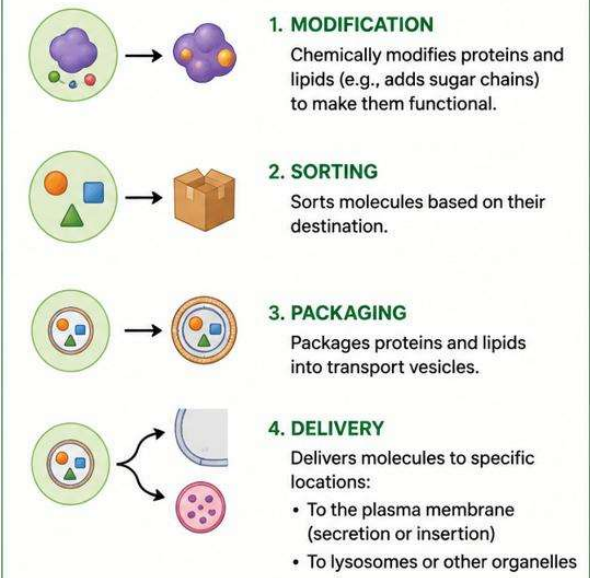
Golgi Bodies

The Golgi apparatus (Golgi bodies) is a membranous organelle that **modifies, sorts, and packages** proteins and lipids for transport to their final destinations inside or outside the cell.

STRUCTURE



FUNCTIONS



In short: The Golgi apparatus acts like the cell's **processing, sorting, and shipping center**.

Figure 2.4: Structure of the Golgi apparatus (Golgi body/Golgi complex).

The Golgi apparatus — also known as the Golgi complex or Golgi body — is named after the Italian physician Camillo Golgi, who first described the organelle. It is found in all eukaryotic cells, both plant and animal, as a membrane-bound structure situated within the cytosol.

Its principal function is the packaging and secretion of proteins. The Golgi apparatus receives proteins from the endoplasmic reticulum, packages them into membrane-bound vesicles, and transports them to their appropriate destinations — including lysosomes, the plasma membrane, or the extracellular space for secretion. It additionally participates in lipid transport and in the formation of lysosomes.

2.7 Other Key Organelles (Overview)

While treated in greater depth in subsequent units, students should note the following organelles as part of the complete eukaryotic organellar inventory:

- Mitochondria: The principal site of ATP generation via aerobic respiration; often termed the "powerhouse of the cell."
- Ribosomes: Sites of protein synthesis; may be free in the cytosol or bound to the rough ER.
- Lysosomes: Membrane-bound vesicles containing hydrolytic enzymes for intracellular digestion and waste breakdown.
- Vacuoles: Storage organelles; especially large and prominent in plant cells, contributing to turgor pressure.
- Centrioles: Cylindrical structures involved in organising the mitotic spindle during cell division (present in animal cells).

Engineering Relevance

Mitochondrial bioenergetics informs the design of biofuel cells and bio-batteries; lysosomal enzyme pathways are exploited in targeted drug-delivery and enzyme-replacement therapies; and the Golgi/ER secretory pathway is central to recombinant protein production in bioprocess engineering.

Unit 3: Tissue Organisation

3.1 What Are Tissues?

A tissue is a group of cells with similar shape and function, forming a level of cellular organisation intermediate between individual cells and organ systems. When functional groups of tissues combine, they form organs — such as the heart, liver, or lungs — each performing a specific role within the organism. The study of tissues is termed histology, and the study of disease-related changes in tissue structure is termed histopathology, an essential tool in medical diagnosis.

3.2 Animal Tissues (Four Types)

Animal bodies are composed of four fundamental tissue types, each with a distinct structural role:

- Epithelial tissue — covers body surfaces and lines internal cavities (e.g., skin, gut lining).
- Connective tissue — supports and binds other tissues together (e.g., bone, blood, cartilage).
- Muscular tissue — responsible for movement, comprising skeletal, smooth, and cardiac muscle.
- Nervous tissue — transmits electrical signals (brain, spinal cord, and peripheral nerves).

3.3 Plant Tissues

Plant tissues are broadly classified into two categories:

- Meristematic tissue — actively dividing cells responsible for growth at root and shoot tips.
- Permanent tissue — differentiated cells performing specific functions, further subdivided into:
 - Simple permanent tissue — parenchyma, collenchyma, and sclerenchyma.
 - Complex permanent tissue — xylem (water transport) and phloem (food transport).

3.4 Hierarchical Organisation of Life

Biological systems are organised in a well-defined hierarchy, each level built upon the one preceding it, producing progressively more complex structures capable of increasingly sophisticated functions:

Cells → Tissues → Organs → Organ Systems → Organism

Unit 4: Comparative Cell Biology — Plant, Animal, and Bacterial Cells

A recurring examination and conceptual theme in cell biology is the systematic comparison of plant, animal, and bacterial cells. The following consolidated tables present these comparisons across structural, biochemical, and physiological criteria.

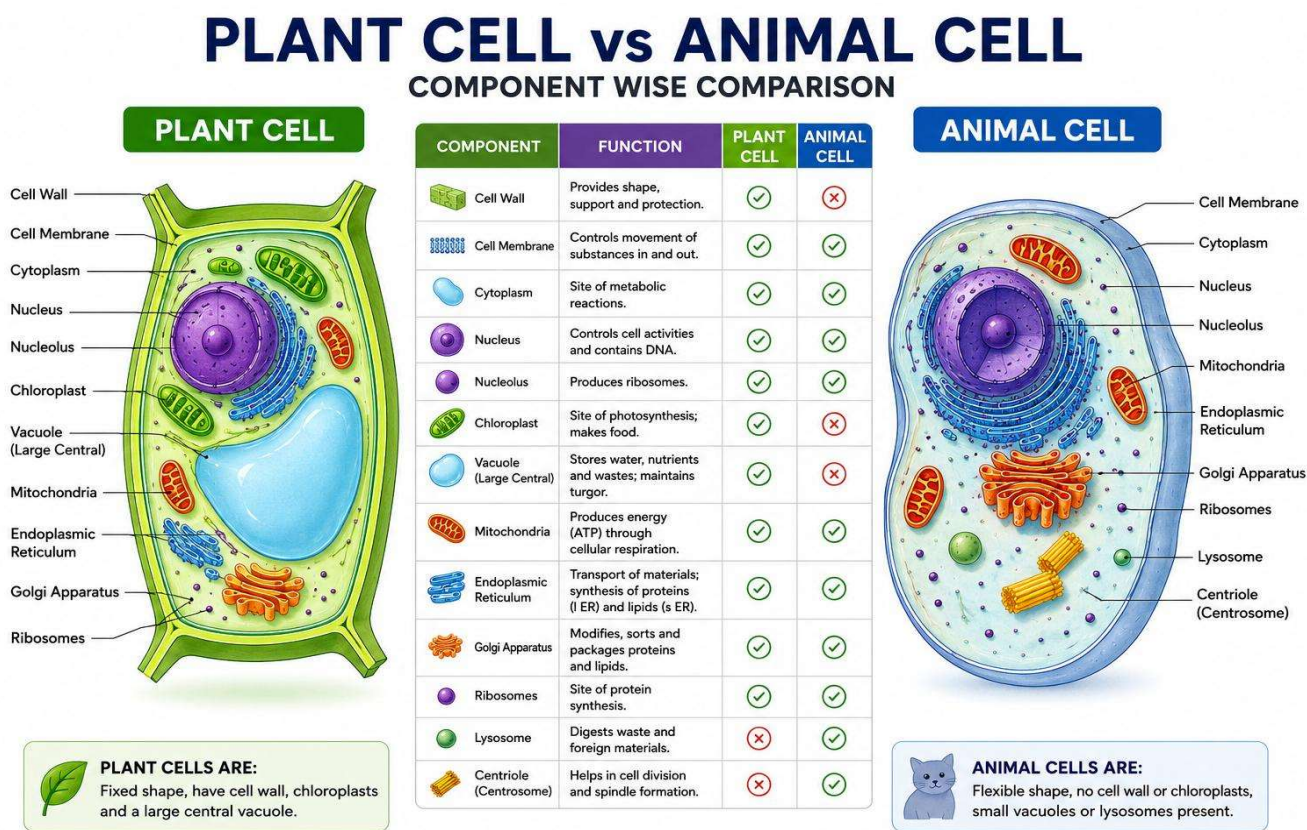


Figure 4.1: Comparative structure of plant and animal cells.

4.1 Structural Comparison

Characteristic	Plant Cell	Animal Cell	Bacterial Cell
Type of cell	Eukaryotic	Eukaryotic	Prokaryotic
Size of cell	10 to 100 μm	10 to 100 μm	0.2 to 2 μm
Shape of cell	Rectangular	Round or oval	Variable — cocci, bacillus, vibrio, spirilla
Cell wall	Present	Absent	Present
Nucleus	Present	Present	Absent (nucleoid instead)

4.2 Biochemical and Organellar Comparison

Characteristic	Plant Cell	Animal Cell	Bacterial Cell
Plasmids	Absent	Absent	Present
Plastids	Absent (except chloroplasts, a related structure)	Absent	Present in some
Mitochondria	Present	Present	Absent
Ribosomes	Larger, 80S	Larger, 80S	Smaller, 70S
Vacuoles	Large, prominent	Small	Large
Golgi apparatus	Present	Present	Absent
Mode of nutrition	Autotrophic	Heterotrophic	Both autotrophic and heterotrophic

4.3 Functional Comparison

Characteristic	Plant Cell	Animal Cell	Bacterial Cell
Lysosomes	Present, but few in number	Present	Absent
Mode of reproduction	Both sexual and asexual	Sexual in higher forms; asexual in lower forms	Both sexual and asexual
Mode of respiration	Aerobic	Aerobic	Both aerobic and anaerobic

Unit 5: Enzymes — Biological Catalysts

5.1 Introduction

Enzymes are biocatalysts that accelerate biochemical reactions within living systems. They are predominantly proteinaceous in nature and exhibit a high degree of substrate specificity. Enzymes function by lowering the activation energy of a reaction, allowing it to proceed more rapidly, without themselves being consumed in the process.

5.2 Structure of Enzymes

Enzymes are biological catalysts composed of one or more long polypeptide chains folded into a specific, well-defined three-dimensional shape. Structurally, a complete functional enzyme (holoenzyme) comprises two components:

- Apoenzyme: The protein component, catalytically inactive when acting alone.
- Cofactor: A non-protein component, which may be a metal ion or an organic molecule such as a vitamin derivative.

A specific region on the enzyme, known as the active site, is where the substrate binds and the catalytic reaction occurs.

Class	Function	Example
Oxidoreductases	Catalyze oxidation-reduction reactions	Dehydrogenase
Transferases	Transfer functional groups between molecules	Transaminase
Hydrolases	Catalyze hydrolysis reactions	Amylase
Lyases	Catalyze addition/removal of groups without hydrolysis	Decarboxylase
Isomerases	Catalyze structural rearrangements	Isomerase
Ligases	Catalyze bond formation coupled with ATP hydrolysis	DNA Ligase

Figure 5.1: Classification of enzymes.

5.3 Characteristics of Enzymes

Property	Description
Catalytic nature	Accelerate biochemical reactions without being consumed
Specificity	Highly specific for particular substrates (Lock-and-Key model / Induced Fit model)
Temperature sensitivity	Optimal activity at a specific temperature (~37°C for human enzymes); high temperatures cause denaturation
pH sensitivity	Optimal activity at a specific pH (e.g., pepsin functions best at pH 2)

Property	Description
Activation energy	Lower the activation energy required for the reaction to proceed
Reversibility	Many enzyme-catalysed reactions are reversible
Cofactors/coenzymes	May require metal ion cofactors (e.g., Zn, Mg) or organic coenzymes (e.g., NAD+)
Inhibition	Susceptible to competitive and non-competitive inhibition

5.4 Mechanism of Enzyme Action

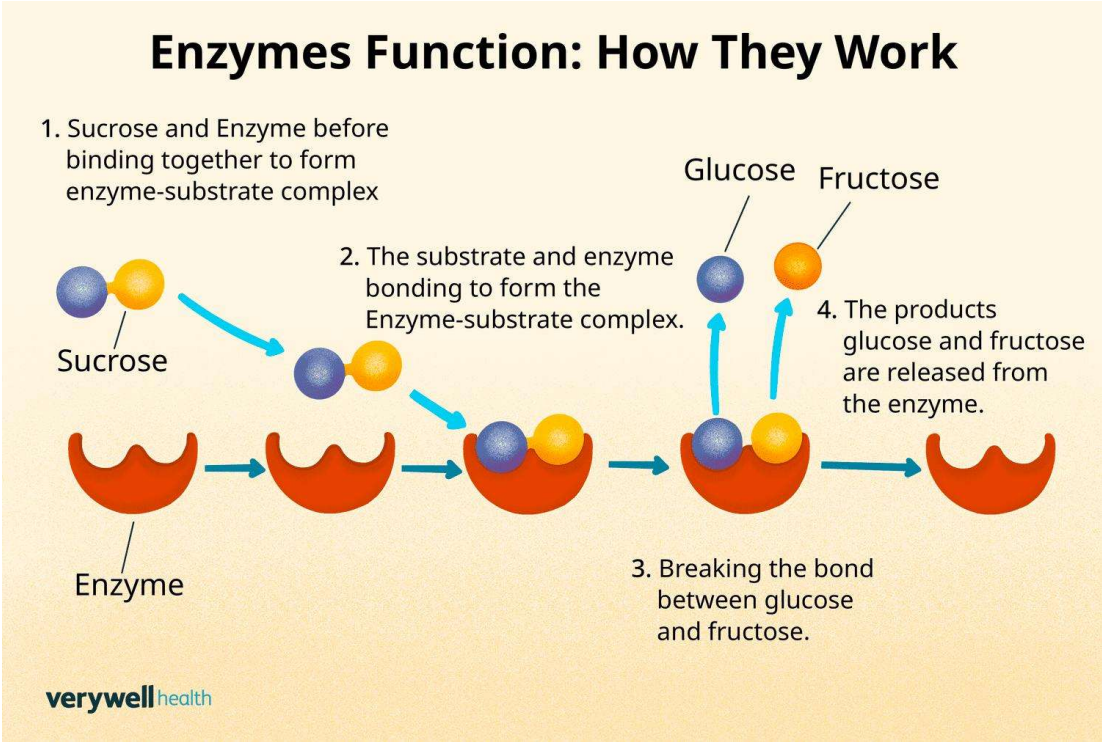


Figure 5.2: Enzyme mechanism — substrate binding, catalysis, and product formation.

Step 1 — Substrate Binding: At the active site, the enzyme binds to a specific substrate molecule, forming an enzyme–substrate complex.

Step 2 — Catalysis: The enzyme facilitates the chemical transformation, either breaking down the substrate (catabolism) or synthesising a new product (anabolism).

Step 3 — Product Formation: The enzyme is released unchanged and is immediately available to catalyse further reaction cycles, while the product(s) is released.

5.5 Physiological Examples

Enzyme	Substrate	Reaction Catalysed
Amylase	Starch	Breaks down starch into maltose

Enzyme	Substrate	Reaction Catalysed
Pepsin	Proteins	Breaks down proteins into smaller peptides
Lipase	Fats	Breaks down fats into glycerol and fatty acids

Engineering Relevance

Enzyme kinetics (Michaelis–Menten behaviour) directly parallels reaction-rate concepts in chemical engineering. Immobilised enzyme systems — enzymes anchored to solid supports — are widely used in biosensors, biofuel production, and industrial bioreactors, and represent an active area of applied nanomaterials and biomaterials research.

Unit 6: Photosynthetic Pigments — Chlorophylls and Carotenoids

6.1 Chlorophylls

Chlorophylls are the green pigments responsible for photosynthesis in plants, algae, and cyanobacteria. Two principal types occur in higher plants:

- Chlorophyll a: The main pigment directly involved in photosynthesis; greenish-blue in colour; absorbs light predominantly in the violet-blue and red regions of the spectrum.
- Chlorophyll b: An accessory pigment that complements chlorophyll a; greenish-yellow in colour; absorbs light in the blue and orange-red regions.

Structure

Chlorophyll molecules contain a porphyrin ring, with a magnesium ion at its centre, together with a long hydrocarbon tail known as the phytol chain. The porphyrin ring is responsible for light absorption, while the phytol tail anchors the molecule within the thylakoid membranes of the chloroplast.

Functions

- Captures solar energy and converts it into chemical energy during photosynthesis.
- Chlorophyll a participates directly in the photochemical reactions of the light-dependent phase.

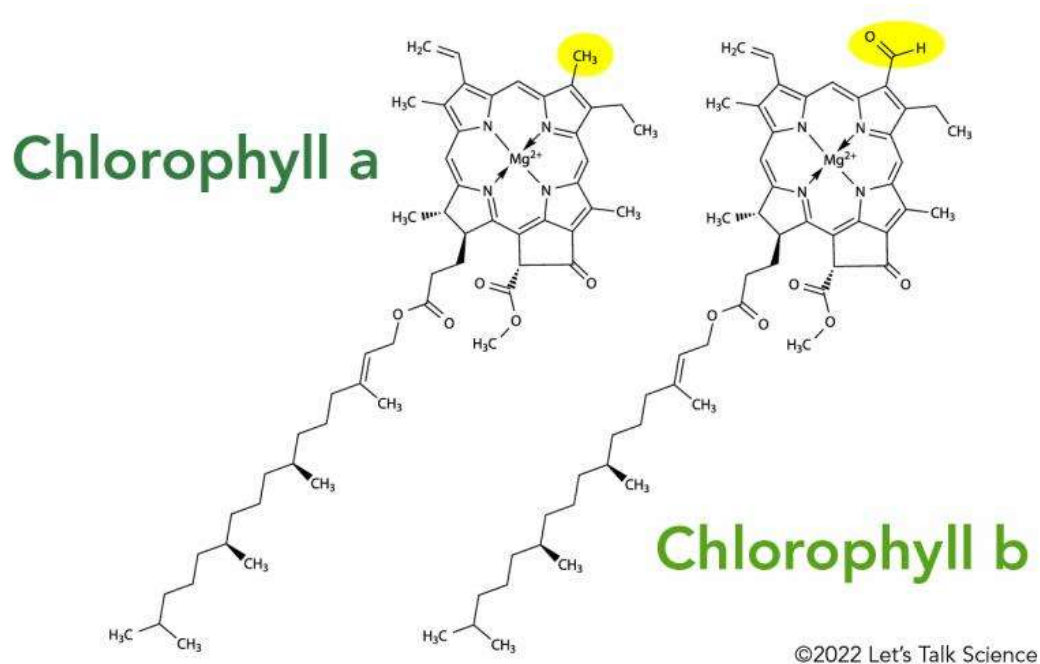


Figure 6.1: Molecular structure of chlorophyll, showing the porphyrin ring and phytol tail.

6.2 Carotenoids

Carotenoids are accessory photosynthetic pigments, typically red, orange, or yellow in colour. They are classified into two principal groups:

- Carotenes: Pure hydrocarbons (e.g., β -carotene), orange in colour.

- Xanthophylls: Oxygen-containing derivatives of carotenes (e.g., lutein), yellow in colour.

Functions

- Act as accessory pigments, absorbing light wavelengths not captured by chlorophylls.
- Protect chlorophyll from photo-oxidative damage by dissipating excess absorbed energy.
- Participate in photoprotection and in the cellular antioxidant defence system.

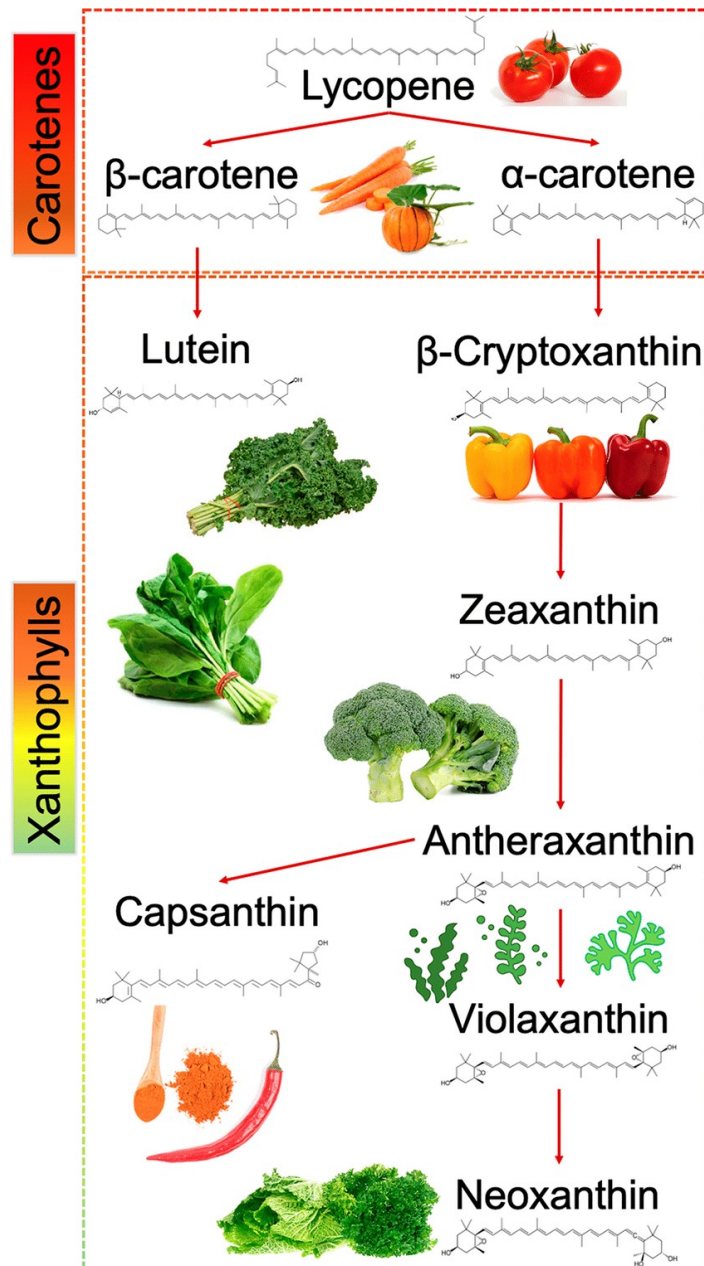


Figure 6.2: Classification of carotenoids.

Engineering Relevance

Chlorophyll- and carotenoid-based light-harvesting mechanisms inform the design of dye-sensitised solar cells and biomimetic photovoltaic materials — an active research direction at the interface of nanomaterials engineering and renewable energy.

Unit Summary — Key Takeaways

- The cell is the fundamental structural and functional unit of life, first observed by Robert Hooke in 1665.
- Cells are classified as prokaryotic (no true nucleus, e.g., bacteria) or eukaryotic (true nucleus, e.g., plants, animals, fungi).
- Major organelles — the cell membrane, cell wall, cytoplasm, nucleus, endoplasmic reticulum, and Golgi apparatus — each perform distinct, essential functions.
- The cell membrane operates according to the Fluid Mosaic Model, providing selective permeability and cell-to-cell communication.
- The nucleus houses the cell's DNA, controls gene expression, and orchestrates cell division.
- Tissues represent an organisational level between cells and organs; animals possess four tissue types, while plants possess meristematic and permanent tissue.
- Plant, animal, and bacterial cells differ systematically in size, wall composition, organelle content, and modes of nutrition, reproduction, and respiration.
- Enzymes are proteinaceous biocatalysts that lower activation energy, exhibit substrate specificity, and are sensitive to temperature and pH.
- Chlorophylls and carotenoids are the principal photosynthetic pigments, absorbing and transferring light energy while protecting the photosynthetic apparatus.
- Cellular and biomolecular principles underpin numerous engineering applications, including drug delivery, biosensors, tissue engineering, bioprocessing, and renewable-energy materials.

Glossary of Key Terms

Term	Definition
Cell	The basic structural and functional unit of all living organisms.
Prokaryote	An organism whose cells lack a true, membrane-bound nucleus.
Eukaryote	An organism whose cells possess a true, membrane-bound nucleus.
Plasma membrane	The selectively permeable phospholipid bilayer enclosing the cell.
Fluid Mosaic Model	The model describing the cell membrane as a dynamic bilayer with mobile embedded proteins.
Cytoplasm	The jelly-like substance filling the cell, comprising the cytosol and suspended organelles.
Cytosol	The liquid component of the cytoplasm, excluding organelles.
Nucleoid	The region in a prokaryotic cell where DNA is freely suspended, without a surrounding membrane.
Nucleus	The membrane-bound organelle containing a eukaryotic cell's genetic material.
Nucleolus	A dense region within the nucleus responsible for ribosome production.
Endoplasmic reticulum (ER)	A network of membranes involved in protein synthesis (rough ER) and lipid synthesis (smooth ER).
Golgi apparatus	An organelle that packages and secretes proteins received from the ER.
Histology	The study of the microscopic structure of tissues.
Meristematic tissue	Actively dividing plant tissue responsible for growth.
Enzyme	A protein biocatalyst that accelerates biochemical reactions without being consumed.
Active site	The specific region of an enzyme where substrate binding and catalysis occur.
Apoenzyme	The protein component of a holoenzyme, inactive without its cofactor.
Cofactor	A non-protein component required for enzyme activity.
Chlorophyll	The primary green pigment responsible for capturing light energy in photosynthesis.
Carotenoid	An accessory photosynthetic pigment providing red, orange, or yellow colouration.

Review Questions

A. Short Answer Questions (2–3 marks each)

- Define a cell and name the scientist credited with its discovery.
- State two structural differences between prokaryotic and eukaryotic cells.
- What is the Fluid Mosaic Model?
- Differentiate between the cytoplasm and the cytosol.
- What is the function of the nucleolus?
- Distinguish between rough and smooth endoplasmic reticulum.
- Name the four types of animal tissue.
- What are apoenzymes and cofactors?
- List two examples of enzymes along with the reactions they catalyse.
- Differentiate chlorophyll a from chlorophyll b.

B. Long Answer / Descriptive Questions (7–10 marks each)

- Describe the structure and functions of the cell membrane, and explain how the Fluid Mosaic Model accounts for its properties.
- With the help of a labelled diagram, explain the structure and functions of the nucleus.
- Compare and contrast plant, animal, and bacterial cells on the basis of structural and biochemical characteristics.
- Explain the mechanism of enzyme action, and discuss the factors affecting enzyme activity.
- Describe the hierarchical organisation of biological systems from cells to organisms, with suitable examples.
- Discuss the structure and biological role of chlorophylls and carotenoids in photosynthesis.

C. Multiple Choice Questions

1. Who is credited with the discovery of the cell?

(a) Robert Brown (b) Robert Hooke (c) Charles Darwin (d) Camillo Golgi

2. Which of the following is absent in a prokaryotic cell?

(a) Ribosomes (b) Cell wall (c) True nucleus (d) Plasma membrane

3. The Golgi apparatus is primarily responsible for:

(a) DNA replication (b) Packaging and secretion of proteins (c) ATP synthesis (d) Photosynthesis

4. Ribosomes in bacterial cells are of which type?

(a) 80S (b) 70S (c) 60S (d) 40S

5. Which pigment is greenish-blue and directly participates in photochemical reactions?

(a) Chlorophyll b (b) Chlorophyll a (c) Xanthophyll (d) Carotene

6. The active site of an enzyme is the region where:

(a) DNA binds (b) The substrate binds and reaction occurs (c) ATP is synthesised (d) Lipids are stored

7. Which organelle is described as the "command centre" of the cell?

(a) Mitochondria (b) Nucleus (c) Golgi apparatus (d) Ribosome

8. Meristematic tissue in plants is responsible for:

(a) Water transport (b) Growth via active cell division (c) Photosynthesis (d) Structural support only

Answer Key

1(b) 2(c) 3(b) 4(b) 5(b) 6(b) 7(b) 8(b)

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